Solution Design Document
Short-VIX Exposure: An ETF-Based Trading System
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1. Executive Business Case & Practical Problem

- **The Challenge:** Inverse volatility ETPs (SVXY & SVIX) provide high long-term returns by exploiting mean reversion within volatility and contango roll yield. However, holding these products creates exposure to significant tail risks in high volatility regimes (e.g.volmageddon in February 2018 that resulted in over 80% losses).
- **The Solution**: We designed a deterministic regime switching allocation strategy (Model 2073). The algorithm uses statistical categorisation of VIX levels and directional transitions in final moments of market sessions to allocate capital between inverse ETPs and cash.
- **Business Value**: In Development, maximum drawdown on synthetic SVXY −1x fell from 87.91% for buy and hold to 22.79% for Model 2073. In the SVIX Final Test, Model 2073 achieved an 8.74% CAGR versus −11.12% for buy and hold, with maximum drawdowns of 22.13% and 79.30%, respectively.

2. High Level Solution Architecture & Pipeline

Raw Market Data →Data Preparation & Cleaning→ Synthetic Data →Regime analysis →VIX State Rules →Execution & Risk Management → Portfolio Output.

3. Repository Structure and Notebooks

Notebook in Repository	Intended Purpose
00_vix_vvix_svxy_data_retrieval.ipynb	Download VIX, VVIX, SVXY, and SVIX data
01_VIX_Futures_CBOE_F1_F2.ipynb	Build the VIX futures F1/F2 series
02_Synthetic_SVXY-ONEx.ipynb	Build synthetic SVXY −1x
03_Synthetic_SVXY-HALFx.ipynb	Build synthetic SVXY −0.5x
04_VIX_Close_Study_1.ipynb	Analyze VIX states and transitions
05_SMA_crossover_trading_system.ipynb	Evaluate the SMA baseline for −1x
06_SMA_crossover_trading_system-HALFx.ipynb	Evaluate the SMA baseline for −0.5x
07-Model_Search.ipynb	Search and select Model 2073
08-Model_2073_SVXY-ONEx.ipynb	Backtest Model 2073 on synthetic SVXY −1x
09-Model_2073_SVXY-HALFx.ipynb	Backtest Model 2073 on synthetic SVXY −0.5x
10-VIX_Sensitivity_Analysis.ipynb	Test sensitivity to small variations of VIX signals near states'borders.
11-Model_2073_Stability_Analysis.ipynb	Evaluate model's stability using sliding windows
12-SVIX_SVXY-ONEx_Regression.ipynb	Compare SVIX and synthetic SVXY −1x returns
13-Model_2073_SVIX_SVXY-ONEx_Comparison.ipynb	Apply Model 2073 to SVIX

- **Data Engineering & Reconstruction (Notebooks 00–03)**: This ingested the key data, VIX,futures,SVXY & SVIX. A pre-event OLS regression estimated the SVXY return used to adjust the February 2018 Volmageddon dislocation in the synthetic −1x and −0.5x series.
- **Regime & Signal Modeling (Notebooks 04-06):** The VIX was divided into five states using four percentile cutoffs (P55, P76, P90 and P96). Analysis was conducted on directions

probabilities between each regime within the 5/15 day timeframes. Baseline strategies were also created with SMA crossovers of 5/87 days (-1x) and 5/21 days (-0.5x).

- **Strategy Selection & Risk Control (Notebooks 07-09):** A grid search was conducted with 2279 iterations to find model 2073.Risk was controlled using regime transitions (S2→ S3,S4→ S3), partial investment (15% of available cash for an initial C2 purchase and 10% for C3, with additional C2 purchases targeting 70% of NAV after costs and additional C3 purchases using 50% of remaining cash), trailing stop loss limits (20%/40% for -1x & 10%/20% for -0.5x) & cash out based on VIX.

4. **Validation & Robustness (Notebooks 10-13):** 200 Monte Carlo noisy tests were performed, and 2328 roll periods of 3 years were tested to ensure that there was consistent reduction in drawdown. The regression of daily SVIX returns on synthetic SVXY −1x returns showed a close relationship over their common history ($R^2 = 0.9866$). Model 2073 was then evaluated on historical SVIX data in the Final Test.
5. **Operational Setup and Managerial Requirements**

- **Execution Process:** Signals and trades use as reference market daily close values. Fractional shares are allowed. Trades include a commission cost of $0.005/share and 0.10% slippage. Cash is held idle in adverse regimes.
- **Environment:** The notebooks run in Python 3.11 and 3.12 environments. Install dependencies from the repository root running the command:

```
python -m pip install -r requirements.txt
```

Then follow the instructions of the README file to run the notebooks.